

The Turboride Voucher API is deployed live on Vercel production and fully conforms to the game developer's reference implementation (voucher-api-reference). Both check and redeem workflows have been validated directly against the production endpoints.

1. Production API Endpoints

POST

https://turbo-ride-contest.vercel.app/vouchers/check

CHECK VOUCHER

POST

https://turbo-ride-contest.vercel.app/vouchers/redeem

REDEEM & START RACE

Both root (/vouchers/...) and API namespace (/api/vouchers/...) are active and SSL-secured on Vercel Edge.

2. Authentication

REQUIRED HTTP HEADER

Authorization: Bearer <API_KEY>

PRODUCTION / LIVE KEY

trb_live_cr_9a87f2e13d4b6c890e51

TESTING / SANDBOX KEY

trb_test_cr_4c81d09e7a2b5f631d82

3. Test Vouchers Dataset (Developer Reference Suite)

Customer	Player ID	Leaderboard Name	Voucher Code	Expected Status
Customer 1 (Reference Suite)	u_5530	Rahul S	7KQ4-M2XD-9PHA	active
			7KQ4-M2XD-9PHB	active
			7KQ4-M2XD-9PHC	active
			7KQ4-M2XD-9PHD	active
			7KQ4-M2XD-9PHE	active
Customer 2 (Reference Suite)	u_8842	Priya K	4RWN-8TJY-3CGF	active
			4RWN-8TJY-3CGH	active
			4RWN-8TJY-3CGJ	active
			4RWN-8TJY-3CGK	active
			4RWN-8TJY-3CGM	active
Used Code	u_5530	Rahul S	USED-2222-2222	used
Expired Code	u_8842	Priya K	EXPD-2222-2222	expired
Void Code	u_8842	Priya K	VOID-2222-2222	void

4. Technical Specifications & Safety Guardrails

- Code Normalization:** Base-32 alphabet excluding ambiguous digits (no 0, O, 1, l, or L). Inputs are stripped of hyphens and spaces and converted to uppercase automatically (e.g. "7kq4 m2xd 9pha" resolves to "7KQ4-M2XD-9PHA").
- Atomic Database Locking:** Redemptions execute inside an atomic row-level update with write lock, guaranteeing zero race conditions on concurrent redemptions.
- Network Idempotency:** Submitting a redeem request with the same session_id returns the identical success payload without error, preventing lost voucher credits on network retries.
- Double-Spend Blocking:** Re-using an already redeemed voucher with a different session_id is immediately rejected with {"ok": false, "status": "used"}.
- Rate Limiting:** Sliding-window rate limiter set to 60 requests per minute with HTTP 429 response upon exceeding threshold.
- Customer Privacy (PII Protection):** The API returns only a sanitized display nickname (first name + last initial, max 20 chars, e.g. "Rahul S"). Customer phone numbers, email addresses, and full names are strictly excluded.
- Financial Safeguards:** A maximum purchase limit of INR 1,00,000 per customer is enforced across checkout and synced via Admin Platform Settings.

5. Live Production Verification Log (Tested on Vercel)

Status	Endpoint / Method	Input Code / Scenario	Verified Result
[PASS]	POST /vouchers/check	7KQ4-M2XD-9PHA	200 OK (status: "active", Rahul S)
[PASS]	POST /vouchers/check	4RWN-8TJY-3CGF	200 OK (status: "active", Priya K)
[PASS]	POST /vouchers/check	USED-2222-2222	200 OK (ok: false, status: "used")
[PASS]	POST /vouchers/check	EXPD-2222-2222	200 OK (ok: false, status: "expired")
[PASS]	POST /vouchers/check	VOID-2222-2222	200 OK (ok: false, status: "void")
[PASS]	POST /vouchers/redeem	Valid Active Code + session_id	200 OK (status: "used", used_at recorded)
[PASS]	POST /vouchers/redeem	Idempotency Retry (Same session)	200 OK (Identical success payload returned)
[PASS]	POST /vouchers/check	Missing / Invalid Bearer Header	401 Unauthorized

6. Quick Connection Summary for Game Developer

- Connection Checklist:**
 - Base URL:** https://turbo-ride-contest.vercel.app
 - Test Key:** trb_test_cr_4c81d09e7a2b5f631d82 (Live Key: trb_live_cr_9a87f2e13d4b6c890e51)
 - The endpoints are deployed, tested, and actively accepting HTTPS requests immediately.